

SOLIU ADEMOLA MUDASHIRU

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RESEARCH SUMMARY

Theorizing, understanding and solving earth system dynamics through data-driven geospatial modelling approaches for informed decision making on water resources management.

Research Interests: Groundwater and hydrogeologic modelling; GIS, remote sensing, and geospatial modelling; Machine learning and AI approaches in geoscience; Environmental geophysics and subsurface characterization; Natural resources assessment and environmental sustainability; Policy and decision making.

EDUCATION

Master of Technology (M.Tech) in Exploration Geophysics (4.43/5.00 or 88.6%)

Federal University of Technology, Akure (FUTA) 2025

Specialization: Groundwater, Environmental, Engineering, Mining

Bachelor of Technology (B.Tech) in Applied Geophysics (3.82/5.00 or 76.4%)

Federal University of Technology, Akure (FUTA) 2018

KEY SKILLS

Programming languages – (Python, R), **GIS** – (ArcGIS Desktop, ArcGIS Online, Google Earth Engine), **Remote Sensing** – (Experience with Sentinel 1 & 2, SRTM, ALOS- PALSAR, LiDAR, and ASTER), **Modelling** – (Hydrologic, Landfill, Ecosystem, Urban green), **Data Analysis** – (Python, R, Excel, PowerBI), **Machine Learning and AI** (Sci-Kit Learn, PyTorch), **Soft Skills** – (Strong writing, teamwork and independent research abilities).

WORK EXPERIENCE

Teaching Assistant – Department of Applied Geophysics

Federal University of Technology, Akure 2026 - Present

- ❖ Assisting in undergraduate instructions and laboratory sessions.
- ❖ Supporting field based geophysical training exercises.
- ❖ Guiding students in data interpretation and technical report writing.
- ❖ Contributing to exam supervision, grading and academic mentoring.

Environmental Scientist – Mukolak GeoConsult

Nigeria Limited, Akure 2023– Present

- ❖ Assisting in the coordination of researchers in geophysical field surveys for groundwater and environmental studies.
- ❖ Supporting on-site measurements of relevant water parameters like EC, pH, water table depth, and discharge measurement.
- ❖ Collaborating in technical report writing.

Research Assistant - Department of Applied Geophysics

Federal University of Technology, Akure

2021 – 2025

- ❖ Conducted groundwater flow modelling and aquifer characterization using GIS-based analytical tools.
- ❖ Interpreted hydrogeological parameters to evaluate groundwater potential.
- ❖ Prepared groundwater monitoring datasets, processed time-series data, and contributed to field data acquisition.
- ❖ Developed maps for groundwater vulnerability, recharge zones, and hydraulic properties using ArcGIS & Google Earth Engine.
- ❖ Mentored three final year undergraduate students in their thesis data collection and writing.

GIS & Remote Sensing Expert – Freelance

Nigeria

2022 – Present

- ❖ Processing multispectral imagery from Sentinel and Landsat tools for land cover mapping and groundwater related studies.
- ❖ Carrying out spatial interpolation and geostatistical analysis using ArcGIS and Python programming language.
- ❖ Producing thematic maps and digital elevation analyses to support environmental assessments

RESEARCH EXPERIENCE**“Artificial intelligence (AI) for land use risk assessment”**

2026

- ❖ Analyzing EO data to determine flood conditioning parameters like runoff generation indicators
- ❖ Employing the effectiveness of Google Earth Engine (GEE) to spatially assess areas of flood occurrence in the study area
- ❖ Evaluating ML algorithms to predict flood vulnerability of the studied region to support effective land use planning

“Machine learning frameworks and data driven modelling for Copper prospectivity”

2025

- ❖ I conceptualized the adopted conditioning factors.
- ❖ I designed the suitable ML algorithms and Entropy model used for the prospectivity prediction.
- ❖ I co-authored the submitted Team slide for the Next Generation Explorers Awards (NGEA) submission.

“Urban groundwater modelling leveraging on Fuzzy logic approach”

2025

- ❖ I designed fuzzy-based hydrologic model integrating factors from remote sensing and field data acquisition.
- ❖ I conceptualized urban water quality indicators (pH, TDS) to predict resource vulnerability.
- ❖ I drafted the accepted PhD proposal from this research.

“Geospatial technology-based MCDA and machine learning algorithms: an ensemble and inter-evaluating frameworks for conceptualizing groundwater potential”

2025

- ❖ I processed and analyzed all the ecological modelling datasets (LULC, DEM, PERSIANN).
- ❖ I performed all the machine learning modelling using Python programming language.
- ❖ I produced the water availability maps based on the models using ArcGIS 10.7 software.

“Conceptualization of MEREC model for resource allocation leveraging on spatially aware geophysical datasets” 2025

- ❖ I coded the Methods Based on Removal Effects of Criteria using Python programming language for the assessment.
- ❖ I conceptualized the spatial geophysical datasets employed as modelling inputs.
- ❖ I produced the resource partitioning map using ArcGIS 10.8 software.

“Prediction of groundwater vulnerability in a basement complex area employing GIS based comparative MCDMs approach” 2024

- ❖ I conceptualized the Multi Criteria Decision Making Models for the groundwater vulnerability assessment.
- ❖ I integrated remote sensing data (i.e. DEM, PERSIANN) with geophysical datasets for the groundwater vulnerability assessment.
- ❖ I wrote the accepted proposal.

FIELD WORK EXPERIENCE**“Hydrogeophysics survey for borehole development”** 2026

- ❖ I set up the R-50 resistivity equipment to acquire the hydrogeophysics data in the study area
- ❖ I analyzed and interpreted the collected horizontal profiling and Schlumberger data with the other team members.
- ❖ I wrote the report detailing the findings.

“Hydrogeologic Modelling Survey” Oba – Ile, Akure, Nigeria 2025

- ❖ I set up the PASSI resistivity equipment to acquire the 1D VES data in the study area
- ❖ I analyzed and interpreted the collected data with the other team members.
- ❖ I wrote the report detailing the findings.

“Borehole Development Geophysical Survey” Awo-Ekiti, Ekiti State, Nigeria 2025

- ❖ I collaborated in the 1D VES data acquisition and the instrumental setup for the data gathering
- ❖ I analyzed and interpreted the collected data with the other team members.
- ❖ I wrote the report detailing the findings.

“Application of MEREC model for Water Analysis” Akure, Nigeria 2025

- ❖ I led the graduate student in the field data collection of well depth and static water level data in the study area
- ❖ I configured the navigation instrument for smooth well location data acquisition.
- ❖ I gave the graduate students insights into the computations of the well depth data

“Iron Ore Deposit Mapping at Ebiya Ajaokuta Local Government”, Kogi state, Nigeria 2023

- ❖ I conducted Vertical Electrical Sounding (VES) data acquisition with the other team members.
- ❖ I interpreted VES results using WinResist software to locate conductive zones diagnostic of iron ore occurrence.
- ❖ I assisted in the curation of magnetic data interpreted to delineate the magnetic ore body from the surrounding geologic formations

“Geophysical survey for subsurface characterization” Igarra, Edo State, Nigeria 2023

- ❖ I interpreted the 2D ERT data acquired using DIPRO™ inversion software to show lateral and vertical visualization of the geologic settings of the study site
 - ❖ I interpreted the ground truth magnetic data to locate geologic contact zones as well as structures that could be inimical to the foundation design of the study site
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PUBLICATIONS AND REPORTS

Published

- ❖ Mogaji, K.A., **Mudashiru, S.A.**, Ozegin, K.O & Oguntade, S.S (2025). Groundwater potential assessment leveraging the hybrid objective model (IDOCRIW-CoCoSo) in the basement terrain of Nigeria: insights from remote sensing and geophysical datasets. *Cleaner Water (Elsevier)*. [Link](#)
- ❖ **Mudashiru, S.A.**, Mogaji, K.A., Ozegin, K.O. (2026). Groundwater vulnerability evaluation using a Python-coded IDOCRIW-MAUT model in heterogeneous geologic environment, Nigeria. *Geosystems and Geoenvironment (Elsevier)*. [Link](#)
- ❖ Ozegin, K.O., **Mudashiru, S.A** & Ejepu, J.S (2026). Geospatial technology-based MCDA and machine learning algorithms: an ensemble and inter-evaluating frameworks for conceptualizing groundwater potential. *Ecological Frontiers (Elsevier)*. [Link](#)
- ❖ Mogaji, K.A., **Mudashiru, S.A.**, Oguntade, S.S & Toyib, M. Y (2026). Geospatial Modelling of Landfill Suitability Using Geophysical and Remote Sensing Data in a Basement Complex Area. *Discover Applied Sciences (Springer Nature)*. [Link](#)
- ❖ **Mudashiru, S.A.**, Olatunji, A.O., Oke, P.A, Adeyanju, O.L., Orewale, T.P (2024). Subsurface competence evaluation using electrical resistivity method at a proposed building site along FUTA staff quarters, Oba Nla, Akure Southwestern Nigeria. *Pakistan Journal of Geology*, 8 (1): 80.87. [Link](#)
- ❖ Mogaji, K.A & **Mudashiru, S.A** (2024). Groundwater potentiality assessment applying python-based IDOCRIW-COCOSO objective modelling algorithm in a multi-faceted geologic environment of Nigeria. *HydroResearch (KeAi & Elsevier)*. [Preprint Online](#)

Under Review

- ❖ Mogaji, K.A., Eval, J.F., Afolabi, D.O., **Mudashiru, S.A.**, Effiong, A.J & Adegoke, Q.O (2025). Landslide susceptibility assessment using Catastrophe Theory and Analytical Hierarchy Process Concepts in Southwestern Nigeria: Insights from geophysical, geological and meteorological data sets. *Discover Geosciences (Springer Nature)*.

Co-Authored Technical Reports

[Link to reports on my github repository](#)

OTHER COMPETENCIES

- “**Critical Writing Expert at Teflon Workshop**” Remote | Nigeria 2025– Present
 - ❖ Developed tailored academic reports, essays, and research-based assignments for diverse clients
 - ❖ Transformed complex topics into insightful documents that showcase high-level of critical reasoning
- “**Project Work Reviewer**”, Federal University of Technology, Akure, Nigeria 2023– Present
 - ❖ I leveraged my research ability to review undergraduate project write up for publications
 - ❖ I offered insights for graduate research work submission into reputable journals
 - ❖ I review graduate students thesis prior to submission to their supervisors
- “**Machine Learning Expert at Infinity Research Consultant**” Remote | UK 2022 – 2024
 - ❖ I developed machine learning models (i.e. XGBoost, Random Forest, SVM etc.) for diverse projects.
 - ❖ I led data analysis projects across several team members.

THESIS

Master's Thesis

“Development of IDOCRIW-COCOSO and IDOCRIW – MAUT Decision Support System Models for Groundwater Potentiality and Vulnerability in a Basement Terrain” 2025

Bachelor's Thesis

“Subsoil evaluation for pre-foundation study using geophysical methods at FUTA staff quarters” 2018

CONFERENCES AND WEBINARS

Second Sub-Saharan African Regional Virtual Conference by SEG	2024
The Role of Geophysics in HDD Webinar by SEG FUTA	2023
Water and Development Symposium Virtual Conference by IHE DELFT	2023
3 rd World Symposium on Climate Change Adaptation (WSCCA- 2017), FUTA	2019

REFERENCES

Available upon request